

$$\begin{array}{ccccccc}
& & & & s' & & \\
& & & & \text{---} & & \\
T[TX, Y] & \xrightarrow{T[TX, \eta_Y]} & T[TX, TY] & \xrightarrow{T[\mu_X, TY]} & T[T^2X, TY] & \xrightarrow{s_{TX, TY}} & [TX, T^2Y] \xrightarrow{[TX, \mu_Y]} [TX, TY] \\
\downarrow s_{X, Y} & & & \searrow & \downarrow T[T\eta_X, TY] & & \downarrow [\eta_X, T^2Y] \\
& & & & T[TX, TY] & & \\
& & & & \searrow s_{X, TY} & & \\
[X, TY] & \xrightarrow{[X, T\eta_Y]} & & & [X, T^2Y] & \xrightarrow{[X, \mu_Y]} & [X, TY] \\
& & & & \downarrow [\eta_X, TY] & & \\
& & & & & &
\end{array}$$

A commutative diagram illustrating a sequence of transformations between various tensor products and brackets. The diagram is organized into two rows of objects, with arrows connecting them.

Top Row Objects (from left to right):
 $T[TX, Y]$, $T[TX, TY]$, $T[T^2X, TY]$, $[TX, T^2Y]$, and $[TX, TY]$.

Bottom Row Objects (from left to right):
 $[X, TY]$, $[X, T^2Y]$, and $[X, TY]$.

Arrows and Labels:

- $T[TX, Y] \xrightarrow{T[TX, \eta_Y]} T[TX, TY]$
- $T[TX, TY] \xrightarrow{T[\mu_X, TY]} T[T^2X, TY]$
- $T[T^2X, TY] \xrightarrow{s_{TX, TY}} [TX, T^2Y]$
- $[TX, T^2Y] \xrightarrow{[TX, \mu_Y]} [TX, TY]$
- $T[TX, TY] \xrightarrow{T[T\eta_X, TY]} T[TX, TY]$ (A self-loop arrow pointing down to a second instance of $T[TX, TY]$.)
- $T[TX, TY] \xrightarrow{s_{X, TY}} [X, T^2Y]$
- $[TX, TY] \xrightarrow{[\eta_X, TY]} [X, TY]$
- $[X, TY] \xrightarrow{[X, T\eta_Y]} [X, T^2Y]$
- $[X, T^2Y] \xrightarrow{[X, \mu_Y]} [X, TY]$

Additional Features:

- A red curved arrow labeled s' connects the top-left $T[TX, Y]$ to the bottom-right $[X, TY]$.
- A double-lined curved arrow connects the bottom-left $[X, TY]$ to the bottom-right $[X, TY]$.